

**Urban Micro-Mobility Demand Forecasting and Feature Importance Analysis: A Case
Study of Seoul Bike Sharing**

Makayla Liang

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Executive Summary

The urban mobility landscape in Seoul is rapidly evolving, with the public bike-sharing system facing the challenge of optimizing fleet distribution to meet fluctuating customer demand. Due to the high sensitivity of bike rentals to external environmental factors and temporal patterns, the system occasionally suffers from regional shortages or oversupply, highlighting the need for a dynamic demand forecasting strategy.

To address these operational challenges, our report focuses on investigating the rental patterns of Seoul bikes and building predictive models to provide valuable insights. A time-series ARIMA model establishes a baseline trend for daily rental volumes, while an ARIMAX exploration tests the inclusion of meteorological factors. Furthermore, a Random Forest model identifies key drivers of rental demand, emphasizing the importance of weather conditions (such as temperature and humidity) and temporal factors.

Combined with our predictive findings, we recommend: implementing dynamic fleet rebalancing strategies during identified peak hours, adjusting maintenance schedules based on weather forecasts, and leveraging promotional campaigns during off-peak seasons to maximize fleet utilization. By implementing these strategies, the Seoul public bike system may better align bike availability with demand, ensuring operational efficiency and customer satisfaction.

Section I: Introduction and Background Information

The public bike-sharing system in Seoul represents a critical component of the city's sustainable transportation infrastructure. As urban populations grow, providing accessible and eco-friendly mobility solutions becomes increasingly important.

After operating in a densely populated metropolitan area, the bike-sharing system faces challenges with unpredictable demand fluctuations. Failing to anticipate these changes results in stations being either entirely empty during peak commuting hours or overcrowded, preventing users from returning bikes.

- **Temporal Demand Volatility:** The rental volume experiences extreme peaks and troughs within a single 24-hour cycle, heavily driven by commuting hours (as illustrated in Appendix 1).
- **Environmental Sensitivity:** Bike rentals are highly susceptible to weather conditions. Factors such as extreme temperatures, heavy rainfall, or snowfall drastically reduce ridership, leading to idle inventory (Appendix 2 & 3).
- **Operational Inefficiencies:** Without accurate forecasting, the manual redistribution of bikes via transport trucks becomes reactive rather than proactive, increasing operational costs and decreasing user reliance on the system (Appendix 4).

To address these challenges, a robust demand forecast is critical for guiding operational decisions. Forecasting provides valuable insights into hourly and daily demand, helping the management team align bike distribution with expected usage. By using predictive modeling as a tool, the Seoul bike-sharing system can navigate weather-related disruptions, manage its physical assets effectively, and enhance the overall user experience.

Section II: Market Position & Operational Context

We utilize an environmental and operational framework to evaluate Seoul Bike's position within the urban micro-mobility sector.

Strengths:

- **High Urban Accessibility & Infrastructure:** Extensive station coverage across major transit hubs provides a strong competitive advantage for daily commuters.
- **Cost-Effective & Eco-Friendly Mobility:** Offers an affordable, zero-emission alternative to cars and buses, appealing to environmentally conscious urban residents.

Weaknesses:

- **Vulnerability to Climatic Extremes:** Heavy dependence on outdoor conditions means severe winters and monsoon seasons drastically reduce ridership.
- **System Maintenance Disruptions:** Scheduled or emergency system shutdowns disrupt continuous usage patterns and complicate baseline forecasting.

Opportunities:

- **Integration of Real-Time Weather Forecasting:** Leveraging short-term meteorological APIs allows for dynamic predictive rebalancing.
- **Growing Urban Sustainability Goals:** Municipal push toward carbon neutrality drives continuous public adoption of micro-mobility.

Threats:

- **Unpredictable Climate Volatility:** Increasing frequency of extreme weather events due to climate change creates severe demand shocks.
- **Alternative Transit Competition:** Expansion of private micro-mobility options and public transit upgrades.

Section III: Data Collected and Methodology

To forecast Seoul Bike's daily rental demand and analyze influencing factors, we utilized the comprehensive Seoul Bike Sharing Dataset (covering 365 days from December 1, 2017, to November 30, 2018, totaling 8,760 hourly records).

Data Granularity and Aggregation:

- **Hourly Panel Data:** The raw dataset records rentals on an hourly basis across 24 hours per day for 365 unique days.
- **Daily Aggregation:** To support time-series modeling, hourly records were aggregated into daily totals (`groupby('Date').sum()`) alongside daily averages for meteorological variables.

Variables Included:

- **Target Variable:** Total daily **Rented Bike Count**.
- **Internal / Operational Factors:** Hour, seasons, holiday indicators, and functioning day status (**Functioning Day**).
- **External Meteorological Factors:** Temperature (°C), Humidity(%), Wind speed (\$m/s\$), Visibility (10m), Solar Radiation (\$MJ/m^2\$), Rainfall(mm), and Snowfall (cm).

Section IV: Feature Selection and Data Manipulation

To identify the primary drivers of bike rentals, we engineered and evaluated features using a Random Forest regression pipeline.

- **Feature Importance Ranking:** The Random Forest model evaluated environmental features (Appendix 7) and demonstrated that Temperature (°C) is by far the single most

dominant predictor of bike-sharing demand, accounting for over 40% of feature importance, followed closely by humidity and solar radiation.

- **Handling Structural Zeros:** Days marked as non-operating (**Functioning Day = 'No'**) were audited to understand system-level shutdowns, ensuring models could distinguish between natural low demand and artificial closures.

Section V: Limitation of Dataset and Model Selection

Working with a 365-day annual micro-mobility dataset introduces specific methodological constraints:

- **Limited Historical Horizon:** A single year of data restricts the model's ability to capture multi-year macroeconomic trends or inter-annual climate anomalies.
- **The ARIMAX Future Exogenous Variable Paradox:** While Random Forest proves temperature is vital, standard time-series models like ARIMAX require future known values for external variables. In long-term forecasting (e.g., 30 days ahead), true future temperatures are unknown; using historical moving averages as proxies flattens the forecast into a conservative horizontal line (Appendix 6).

To address these constraints, we paired a baseline ARIMA model for temporal trend extrapolation with a Random Forest model for feature attribution.

Section VI: Time Series Model (ARIMA vs. ARIMAX) & Critical Insights

We implemented a standard univariate $\text{ARIMA}(1,1,1)$ model to forecast daily rental trends without relying on flawed future proxy variables.

- **Baseline ARIMA Performance:** By relying solely on historical autoregressive patterns (Appendix 5), the model captures baseline demand trends while outputting a 95% confidence interval that widens over time to reflect increasing uncertainty.
- **The ARIMAX Trade-off:** Our comparative analysis yields a critical methodological insight: Although Random Forest confirms that temperature is a primary demand driver, ARIMAX is ill-suited for unassisted long-term forecasting because future meteorological inputs are unavailable. Consequently, weather-based exogenous variables are optimally deployed in short-term operational windows (1 to 7 days) backed by verified weather forecasts, whereas long-term strategic planning relies on baseline temporal trends.

Section VII: Random Forest Model Feature Analysis

The Random Forest model successfully uncovered the nonlinear relationships governing urban bike rentals:

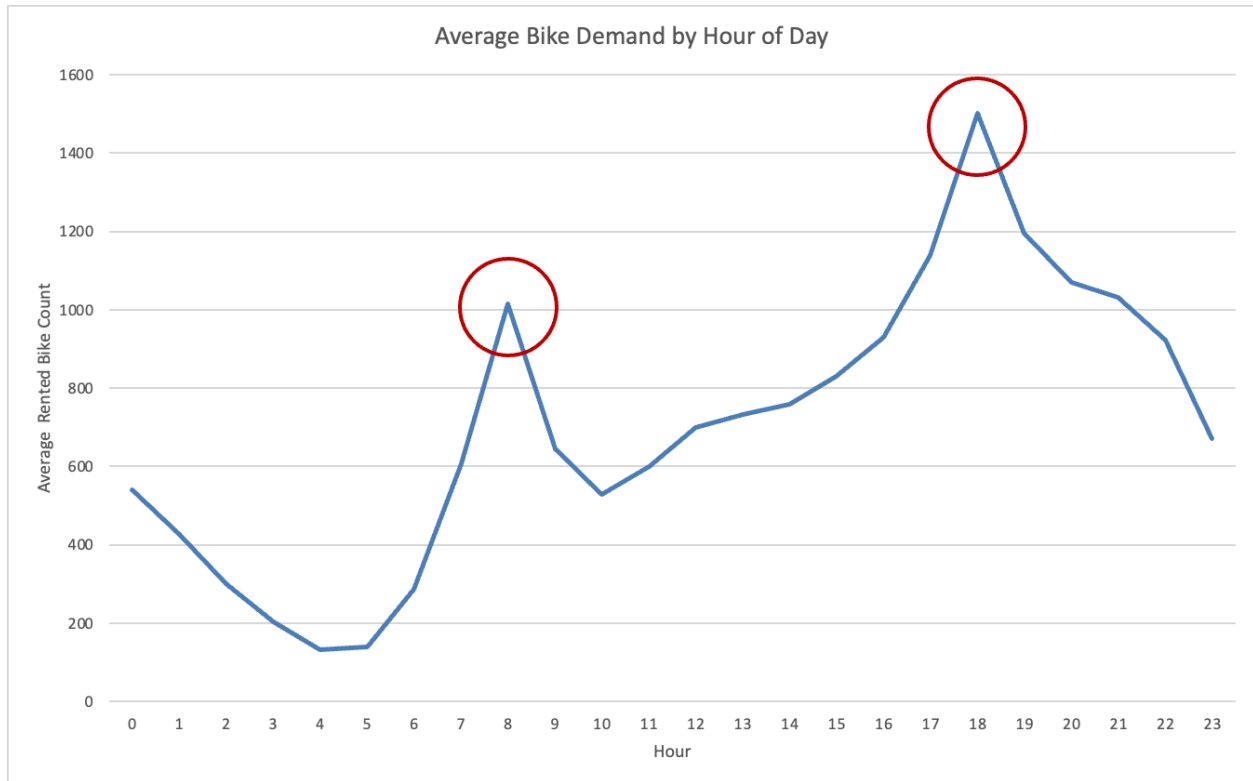
1. **Temperature Dominance:** Thermal comfort dictates micro-mobility adoption; ridership scales proportionally with favorable temperatures up to an optimal threshold (Appendix 2).
2. **Precipitation & Humidity Constraints:** Rainfall and excessive humidity act as immediate deterrents, causing sharp drops in daily rentals (Appendix 3).
3. **Operational Stability:** Factoring in operational status ensures that predictive errors caused by system maintenance days are properly isolated.

Section VIII: Recommendations & Implementation

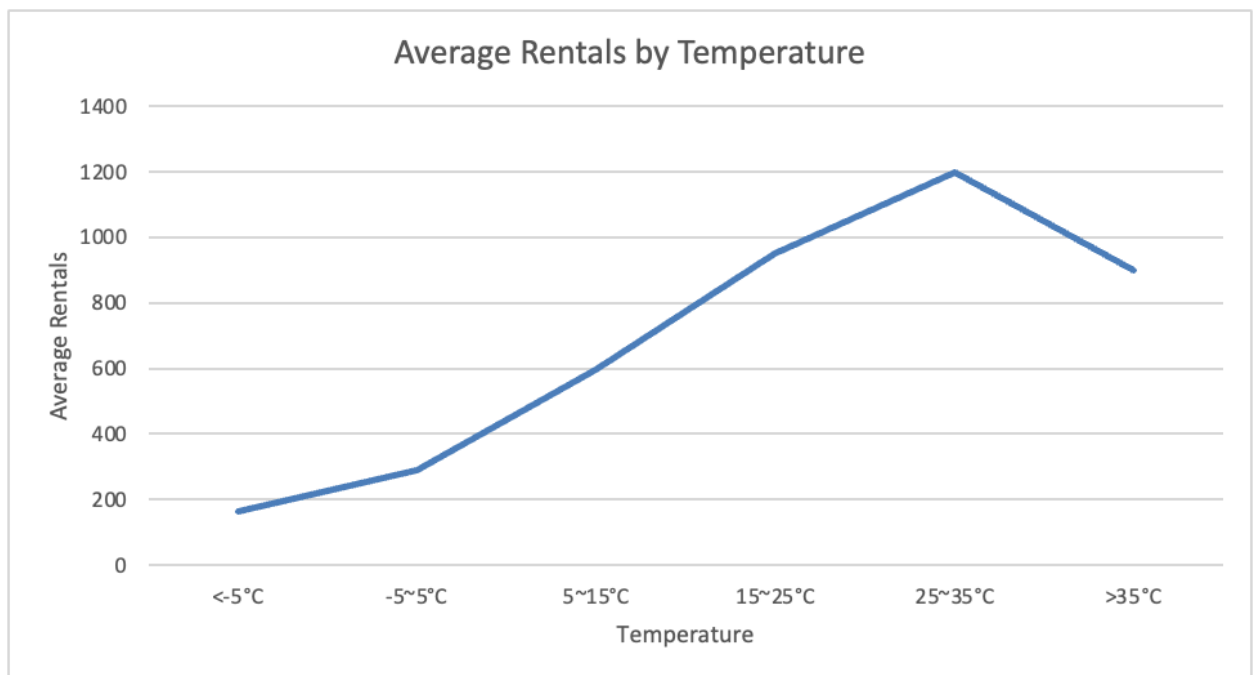
Based on our empirical findings, we propose three strategic recommendations for Seoul Bike management:

1. **Deploy Short-Term ARIMAX for Dynamic Fleet Rebalancing:** Transition from long-term forecasting to 3-to-7-day operational windows powered by ARIMAX and live weather API inputs to predict exact station surpluses and deficits.
2. **Prioritize Temperature-Triggered Maintenance Scheduling:** Use thermal thresholds identified by the Random Forest model to schedule routine fleet maintenance during expected low-demand cold or rainy periods.
3. **Incorporate System Shutdown Flags in Planning:** Explicitly model Functioning Day parameters to prevent maintenance closures from skewing baseline demand metrics.

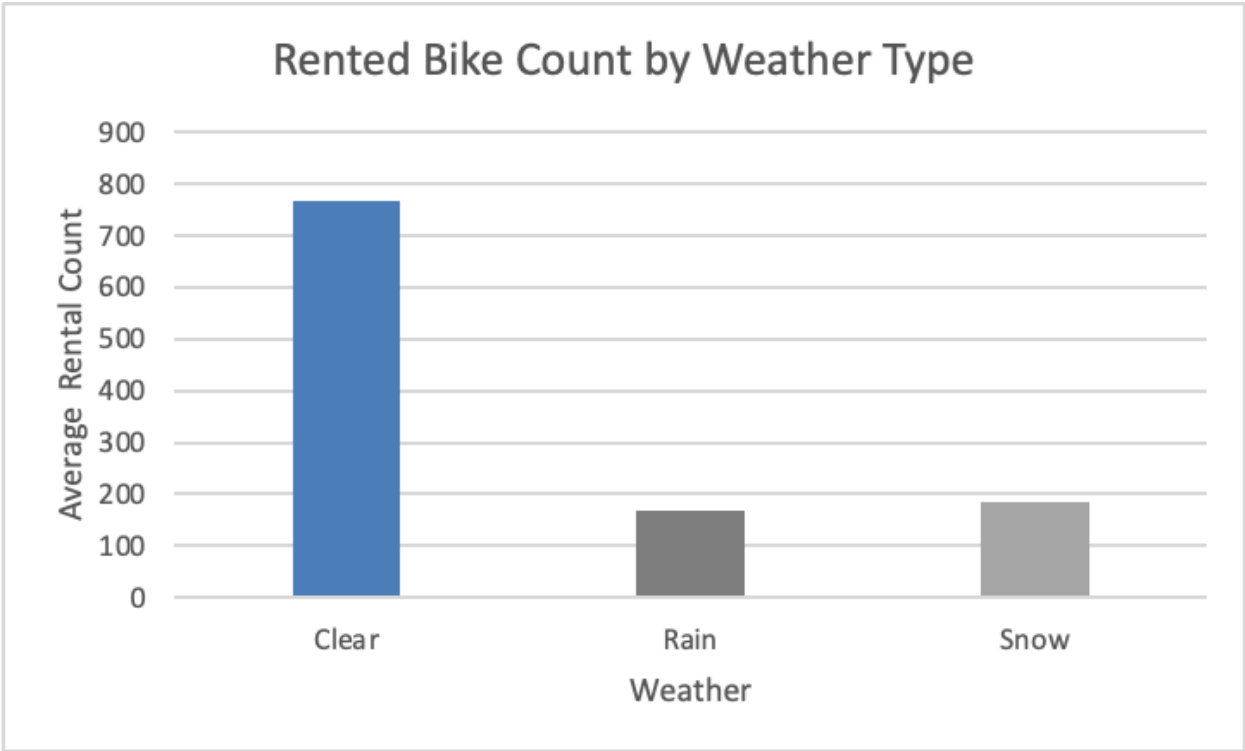
Appendix 1 - Business Problem : Temporal Demand Volatility



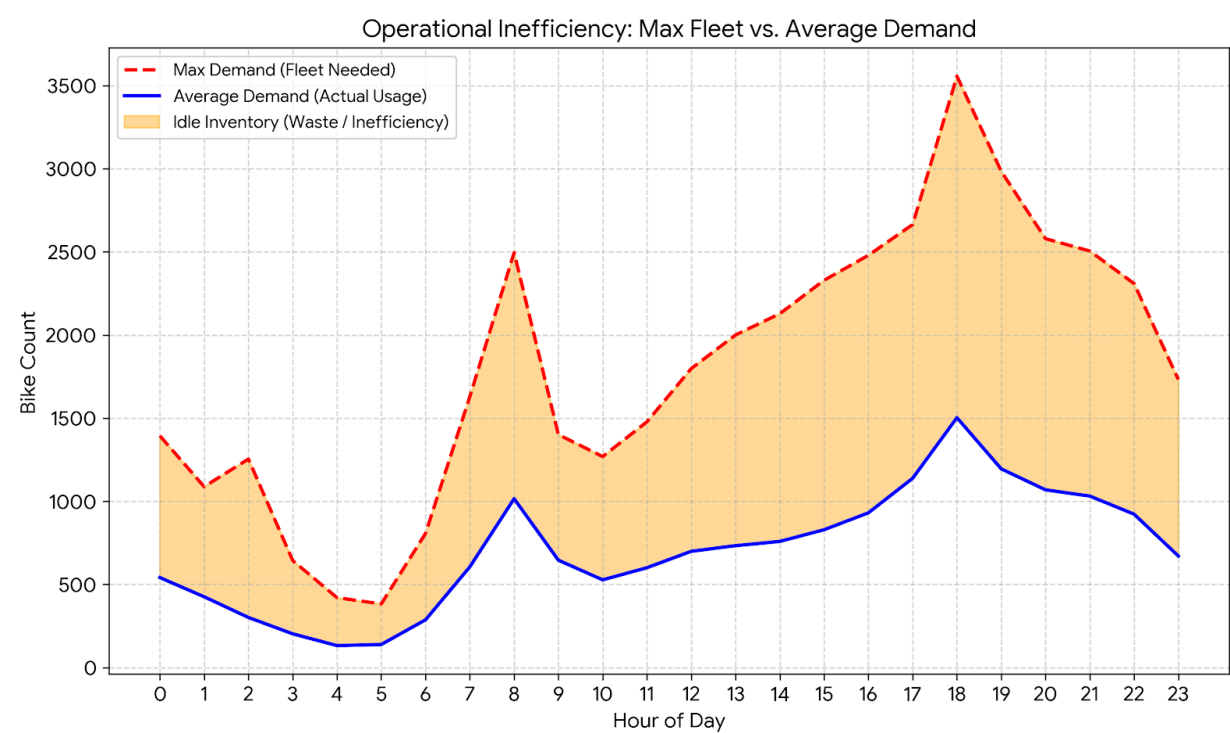
Appendix 2 - Business Problem : Temperature Non-linear Impact



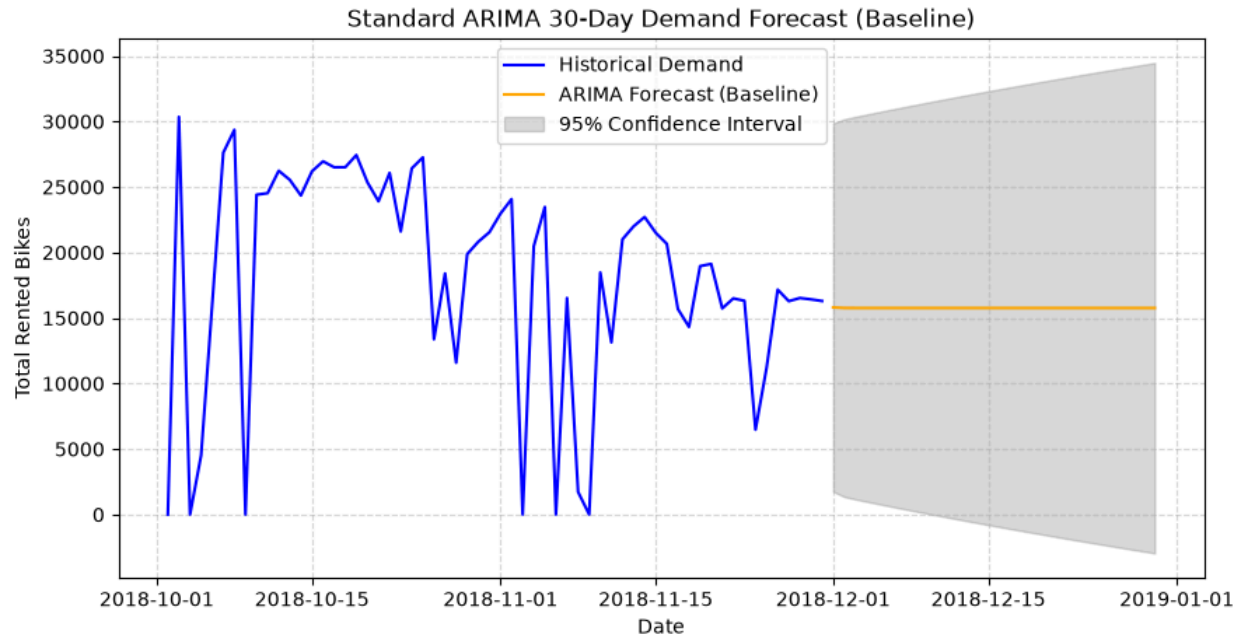
Appendix 3 - Business Problem : Weather Condition Impact



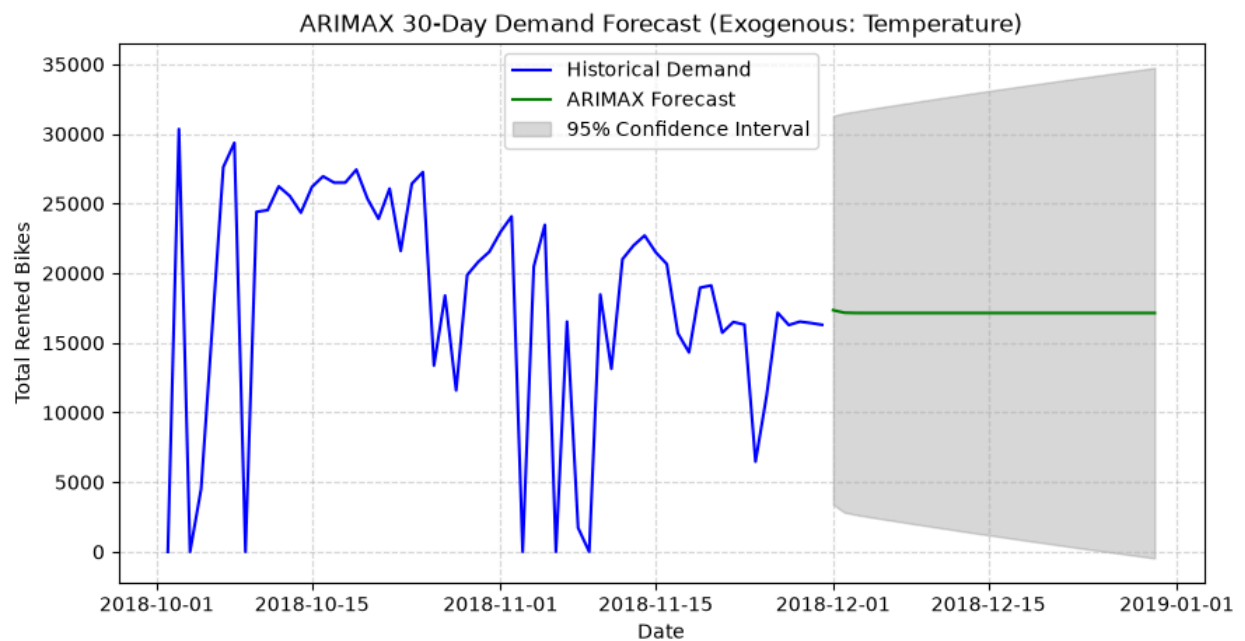
Appendix 4 - Business Problem : Operational Inefficiencies



Appendix 5 - Sales Forecast : ARIMA



Appendix 6 - Sales Forecast : ARIMAX



Appendix 7 - Feature Selection :Random Forest

